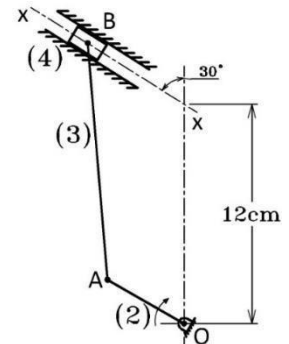


MDP 212 Mechanics of Machines

Question (1): (10 points)

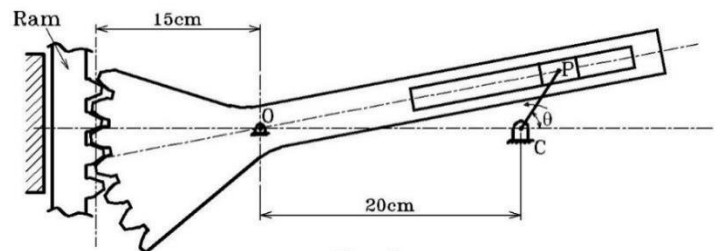
In the slider-crank mechanism shown in figure, the slider B travels along the centerline xx , $OA = 4$ cm and $AB = 15$ cm. Name the finite and infinite cranks.

- Find the two extreme positions of B.
- Determine the stroke.
- Determine the ratio between forward and return times.



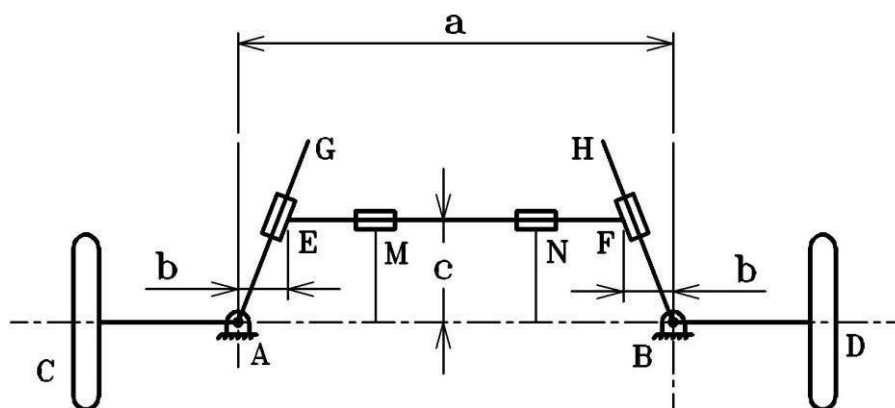
Question (2): (15 points)

The figure shows the quick-return mechanism of a slotting machine. The toothed sector gears with a rack on the ram which carries the toolbox. If the ratio of the times taken for cutting and the return strokes is to be 1.5: 1. Determine the length of the driving crank CP. Determine the stroke, the cutting and the idle angles.



Question (3): (15 points)

In the Davis steering-gear shown in Figure.23, the stub axles CA and DB are pivoted at A and B on the rigid axle and are continuous with the arms AG and BH respectively. The cross-link EF is pivoted at the ends to the blocks sliding on the arms AG, BH and slides in the guides at M & N. Show that correct steering is obtained if the distance between front and rear axles is $(ac/2b)$.



Question (4): (15 points)

Design a four-bar mechanism that converts continuous rotation into oscillating motion (crank rocker mechanism). Given that the crank length $r_2 = 100$ mm, coupler length $r_3 = 200$ mm and follower length $r_4 = 300$ mm.

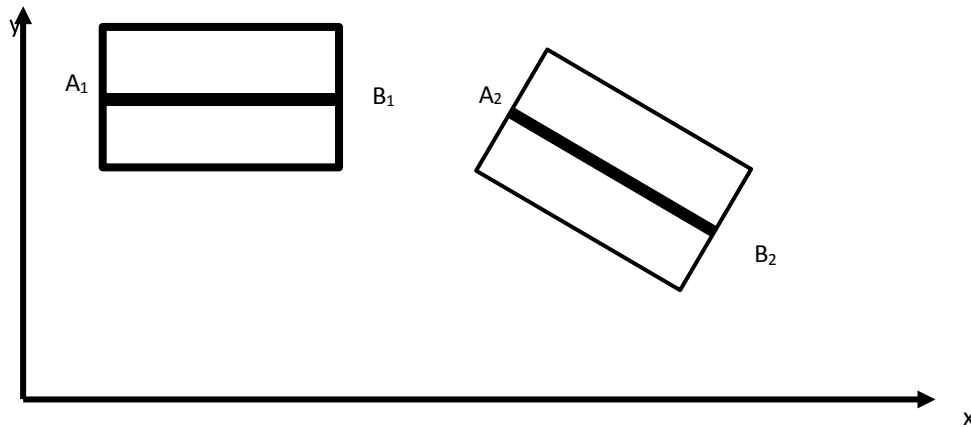
- Find the range of the fixed link r_1 to get a crank rocker linkage.
- Determine the specific value of r_1 such that the transmission angle is more than of 45° .
- Calculate the stroke of the follower and determine the time ratio of the linkage.

Question (5): (15 points)

Design a slider crank mechanism with quick return ratio 1:1.25. The offset is 4 cm. The length of the stroke is 10 cm. Determine the length of the crank and the length of the connecting rod.

Question (6): (15 points)

The Design a four-bar mechanism to move the shown object from position 1 to position 2, where the ratio between the forward time to return time is 1.25: 1. Consider the output is the rocker.



Question (7): (15 points)

Design a four-bar mechanism to move link AB, 5cm length, between the three shown positions with equal time. Design another four-bar mechanism to drive the rocking mechanism, considering that the rocking hinges should be O and Q

